

Simulation and Experimental Characterization of a Mach–Zehnder Interferometer on 220 nm SOI

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Index Terms—Mach-Zehnder interferometer, silicon photonics.

I. INTRODUCTION

Mach–Zehnder interferometers (MZIs) are fundamental components in photonic integrated circuits, with applications in modulation, filtering, and sensing. In an imbalanced MZI, the wavelength-dependent interference spectrum provides a direct method for determining the waveguide group index, n_g , from the free spectral range (FSR).

In this work, MZIs are designed for fabrication on a 220-nm silicon-on-insulator (SOI) platform. Waveguide properties are modelled using Ansys Lumerical MODE, and the MZI spectral response is simulated using INTERCONNECT. The simulated results are compared with experimental transmission spectra to extract n_g and evaluate agreement between simulation and measurement.

II. THEORY

An imbalanced Mach–Zehnder interferometer (MZI), shown in Fig. 1, splits an input optical field into two arms of unequal length. Upon recombination, the accumulated phase difference between the two paths produces a periodic wavelength-dependent transmission spectrum.

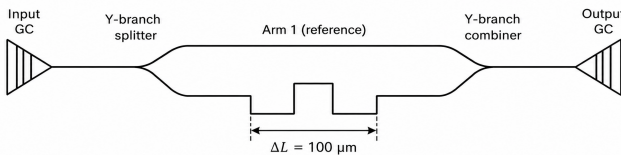


Fig. 1. Schematic of an imbalanced Mach–Zehnder interferometer with path-length difference ΔL .

The phase difference $\Delta\phi$ between the two arms is

$$\Delta\phi = \beta\Delta L, \quad (1)$$

where β is the propagation constant, and ΔL is the path-length difference between the two arms.

The periodicity of the interference spectrum is characterized by the free spectral range (FSR), defined as the wavelength separation between adjacent transmission maxima or minima. For a small wavelength range, the FSR is approximately

$$\text{FSR} \approx \frac{\lambda^2}{n_g \Delta L}, \quad (2)$$

where λ is the free-space wavelength and n_g is the group index of the guided mode. The group index can therefore be extracted from the measured MZI spectrum as

$$n_g \approx \frac{\lambda^2}{\text{FSR} \Delta L}. \quad (3)$$

This relationship provides the basis for comparing the experimentally extracted group index with the simulated waveguide group index.

III. MODELLING AND SIMULATION

A. Waveguide Model

A 500 nm $\times$ 220 nm silicon strip waveguide with SiO₂ cladding was modelled using the finite-difference eigenmode (FDE) solver in Ansys Lumerical MODE. The fundamental TE mode was simulated over a wavelength range of 1500–1600 nm, with a reference wavelength of $\lambda_0 = 1550$ nm.

The wavelength-dependent effective index obtained from the simulation was fitted to the compact model

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (4)$$

where $n_1 = 2.45$, $n_2 = -1.13$, and $n_3 = -0.04$, and wavelength is expressed in μm . The group index is related to the effective index by

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (5)$$

At 1550 nm, the simulated group index was $n_g = 4.177$.

Figure 2 shows the simulated electric-field intensity of the fundamental TE mode.

The simulated effective index and group index over the wavelength range of interest are shown in Figs. 3 and 4.

B. MZI Model

An imbalanced MZI with a path-length difference of $\Delta L = 100 \mu\text{m}$ was modelled. The wavelength-dependent transmission of an ideal MZI is described by

$$T(\lambda) = \frac{1}{2} \left[1 + \cos \left(\frac{2\pi n_{\text{eff}}(\lambda) \Delta L}{\lambda} \right) \right], \quad (6)$$

where $T(\lambda)$ is the normalized transmission and $n_{\text{eff}}(\lambda)$ is obtained from the waveguide model.

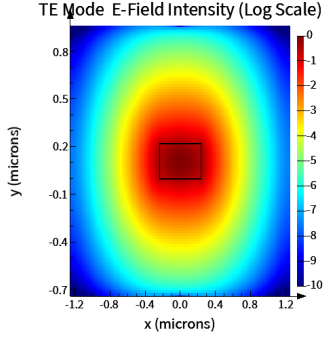


Fig. 2. Simulated electric-field intensity of the fundamental TE mode in the 500 nm $\times$ 220 nm silicon strip waveguide (logarithmic scale).

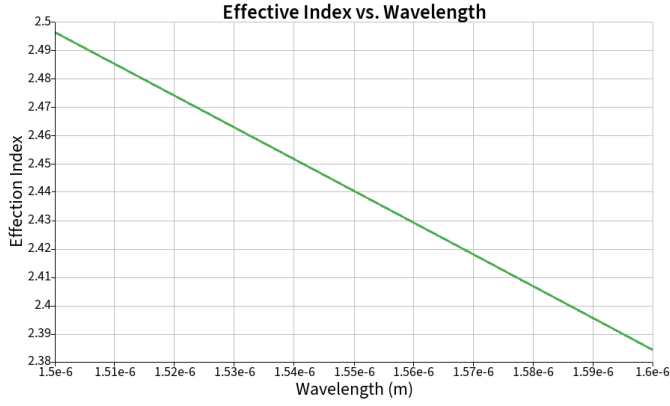


Fig. 3. Effective index n_{eff} versus wavelength for the fundamental TE mode.

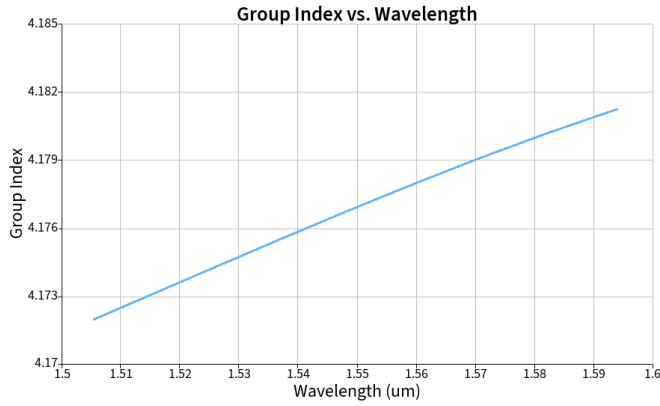


Fig. 4. Group index n_g versus wavelength for the fundamental TE mode.

The MZI circuit was simulated in Ansys Lumerical INTERCONNECT over the wavelength range 1500–1600 nm. The physical layout of the $\Delta L = 100 \mu\text{m}$ MZI is shown in Fig. 5.

C. Simulation Results

The simulated transmission spectrum of the $\Delta L = 100 \mu\text{m}$ MZI is shown in Fig. 6. The periodic response results from the wavelength-dependent phase difference accumulated between the two MZI arms.

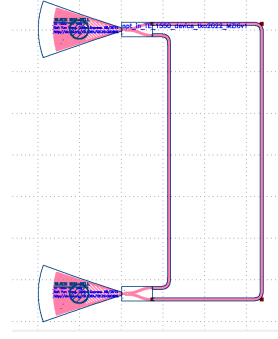


Fig. 5. KLayout layout of the imbalanced MZI with $\Delta L = 100 \mu\text{m}$.

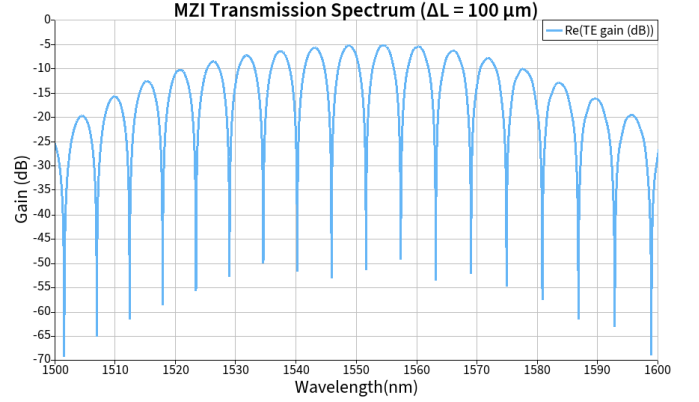


Fig. 6. Simulated transmission spectrum of the $\Delta L = 100 \mu\text{m}$ MZI over 1500–1600 nm.

The simulated FSR near 1550 nm was compared with the value predicted from the waveguide group index using

$$\text{FSR} \approx \frac{\lambda^2}{n_g \Delta L}. \quad (7)$$

Using $n_g = 4.18$ at $\lambda = 1550 \text{ nm}$ and $\Delta L = 100 \mu\text{m}$ gives a theoretical FSR of approximately 5.75 nm.

TABLE I
MZI SIMULATION RESULTS NEAR 1550 NM

ΔL (μm)	n_g	Theoretical FSR (nm)	Simulated FSR (nm)
100	4.18	5.75	5.70

The simulated average FSR was 5.70 nm, in close agreement with the theoretical value of 5.75 nm calculated using the MODE group index of $n_g = 4.18$. The group index extracted from the INTERCONNECT spectrum was $n_g = 4.22$, demonstrating good agreement between the waveguide and circuit simulations.

IV. FABRICATION

To be completed later.

V. EXPERIMENTAL DATA

To be completed later.

VI. ANALYSIS

To be completed later.

VII. CONCLUSION

To be completed later.